TRIPLET 03 · TSYP14 · CITY BRAIN CONVERGENCE

One city.
Three layers.
Zero blind trust.

Twin what is happening?

City Brain what should we do?

Guardian is it true?

THE PROBLEM A smart city that acts on raw sensor data can be **spoofed, replayed or flooded** — and send the wrong unit, or open the wrong valve.

THE TRIPLET · HOW THE LAYERS TALK

Every reading is judged. Every action is audited.

01 · GROUND TRUTH

Twin

What is really happening?

city engine at 10 Hz
signs every reading
checks every call it receives

python · fastapi · three.js

signed readings · HMAC + seq · 1/s

containment · audited calls

03 · TRUST

Guardian

Is this data telling the truth?

7 checks · 9 attack patterns

signed verdict

actuator calls + trust_ref

02 · DECISION

City Brain

What should we do?

rules · agents · solvers · 1 gate

● LAYER 1 · THE TWIN

The only layer that knows the truth.

A real-time city: cars, walkers, signals, fires and floods. Partners only see it through **signed sensors** — and nothing moves without a call.

- ▸ IDM · MOBIL traffic
- ▸ social-force crowds
- ▸ LLM teacher → XGBoost student
- ▸ HMAC-SHA256 + seq_no

10Hz
whole city ticks, one laptop, no cloud

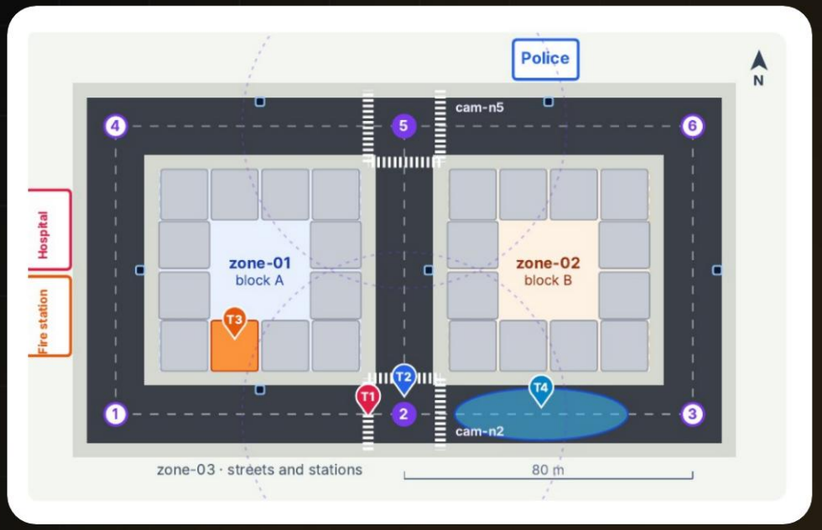
397
tests · 303 Python + 94 Vitest

4/4
scenarios PASS end to end

TWIN TEAM SPEAKING

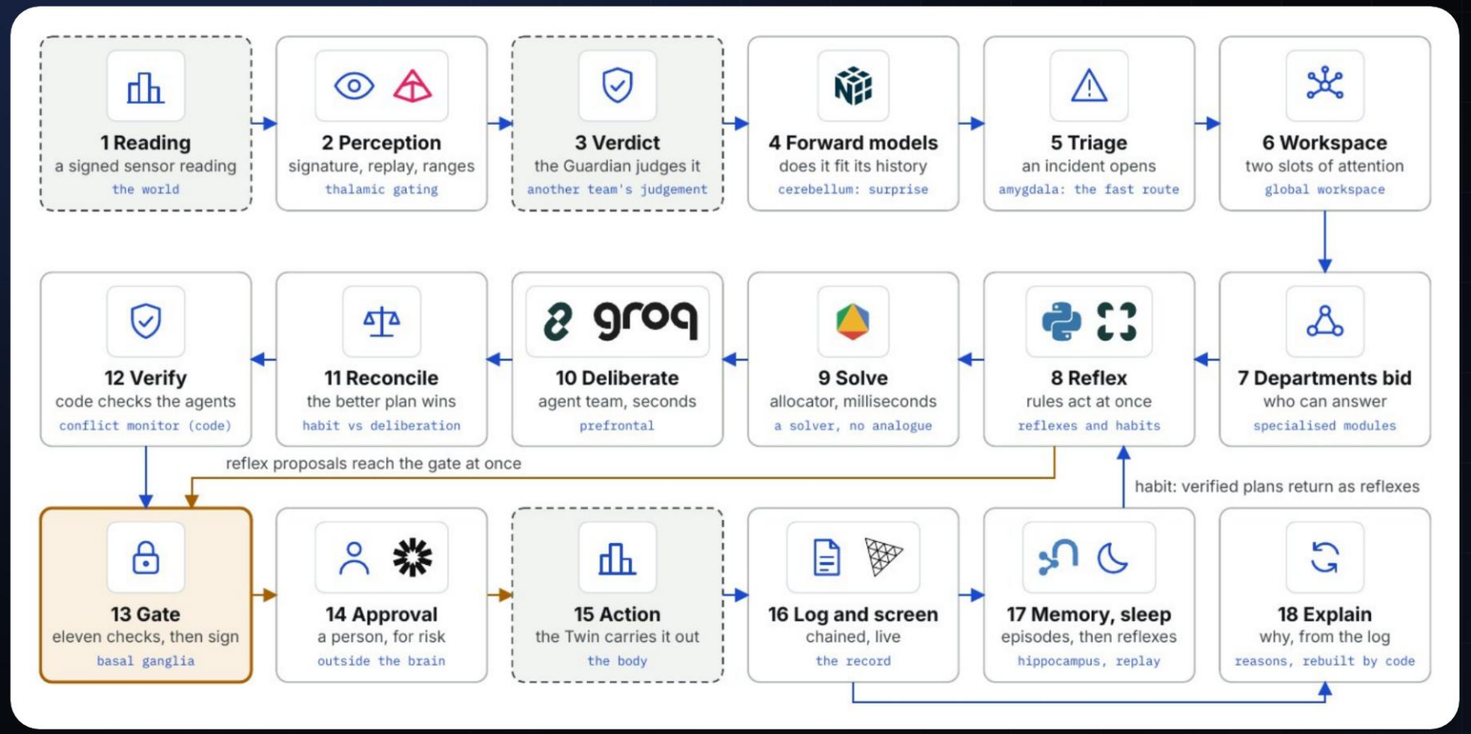
live 3D viewer · two blocks, 24 buildings, 3 emergency units





Fast reflex, thorough plan — one way out.

reading → perception → verdict → reflex / agents (LangGraph · Groq) / solvers (OR-Tools) → gate → action



15ms

reflex dispatch after the evidence — the agent team's solver-checked plan follows through the same gate

11

checks in the safety gate · trust_ref required on every call

24/24

red-team attacks refused · 1,114 tests

● LAYER 3 · THE GUARDIAN

GUARDIAN TEAM SPEAKING

Is this data telling the truth?

A fake flood at the port: `1.80 m` reported, `0.40 m` real, no rain.

L1	L2	L3a	L3b	L4	L5	L6
HMAC	fresh	range	city	dynamics	corrob.	alive

UNTRUSTED

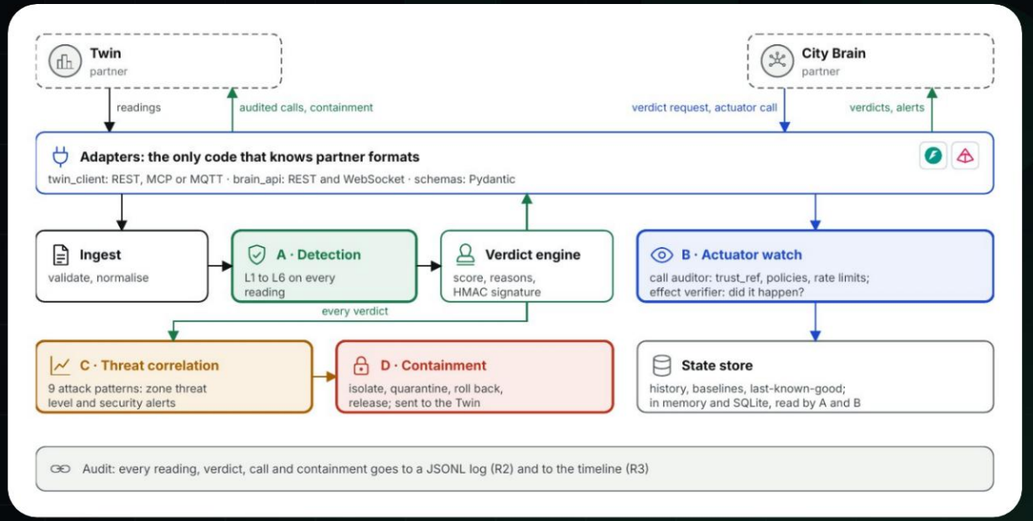
< 10 s · sensor isolated · rollback serves 0.41 m · valve untouched

9
attack patterns correlated across readings

11/11
scenarios pass · logs committed

95%
line coverage · 111 tests

detection · actuator watch · threat correlation · containment



Simple, explainable detectors over a black box: every verdict is **signed** and names its **reasons**. Missing verdict? No action.

RESILIENCE · WHEN SOMETHING BREAKS

Break any piece. The city fails safe.

Missing, expired or untrusted verdict → `unknown`, never `trusted`. Every refusal is logged with a reason.

FAILURE

The Guardian stops answering

Brain acts on **nothing unjudged**. The Twin keeps running.

T15

FAILURE

The Twin goes silent

Every sensor marked **unknown**, never trusted. Health degraded.

T7

FAILURE

A forged or replayed reading

Marked **untrusted**; the Brain refuses it with a code.

T3 · 24/24 red-team

FAILURE

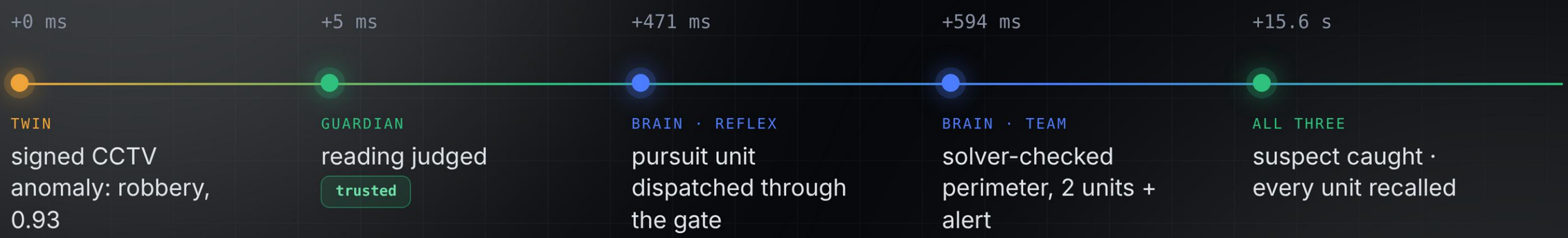
A device is quarantined

Guardian **refuses the call**; the Brain sends another station.

T6 · T16

TOGETHER · SCENARIO T14 · A ROBBERY AND A PURSUIT

From camera to arrest, nothing acts on unjudged data.



No verdict, no action.

Each layer runs alone with mocks.

Partners are config, not code.

One trigger, one log.

Live demo →